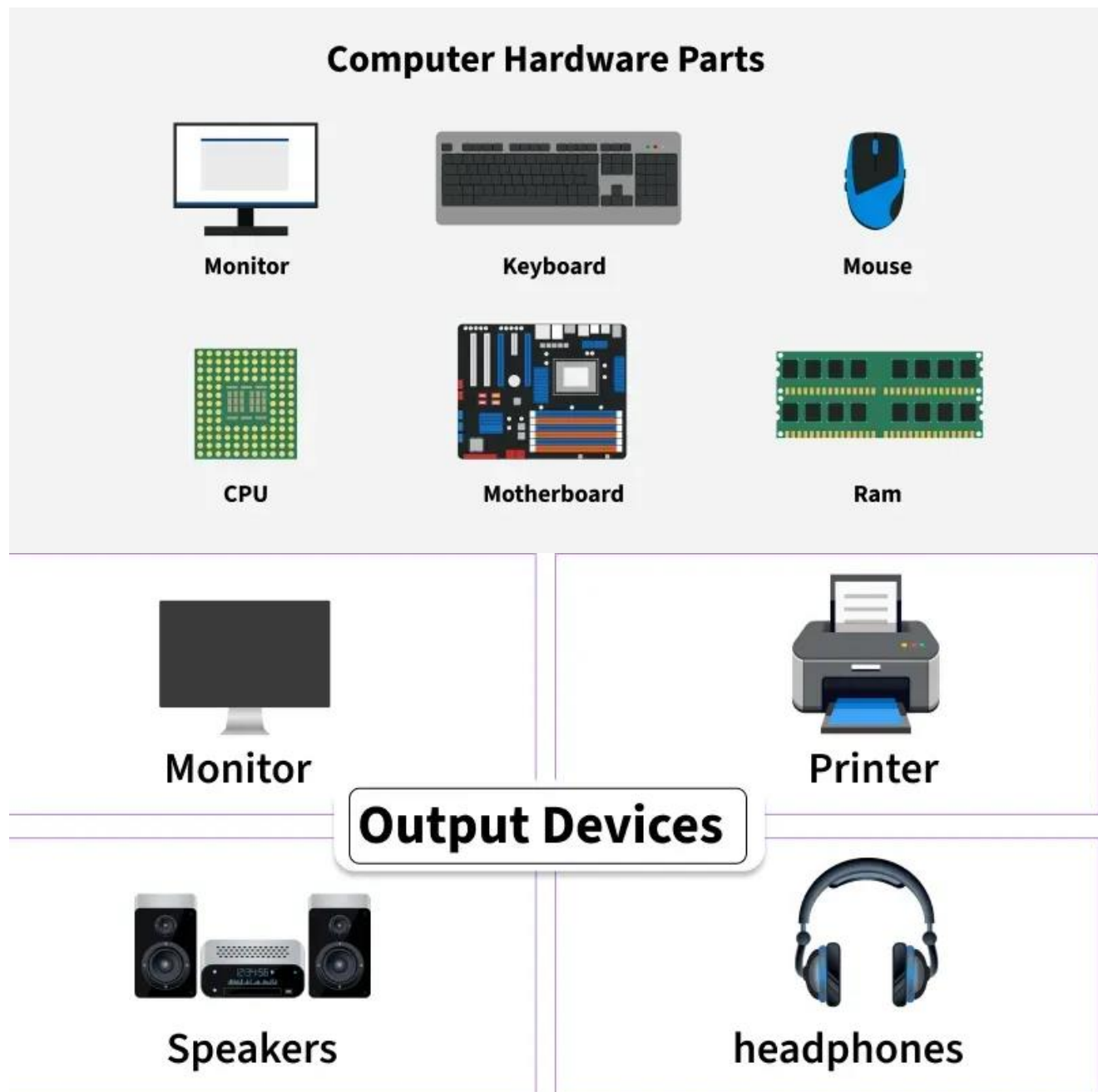


1.Computer Hardware

Computer hardware refers to the **physical components** of a computer that you can see and touch. These components work together to process input and deliver output based on user instructions. In this article, we'll explore the different types of computer hardware, their functions, and how they interact to make your computer work.



2. Materials and Equipment

Hardware Components	Networking & Tools
ATX Motherboard	8-port Ethernet Switch
CPU (Intel Core i5 / AMD Ryzen 5)	Cat5e Ethernet Cables (x6)
DDR4/DDR5 RAM (2 x 8 GB)	6 PCs with NICs (Windows 10/11)
500W ATX Power Supply (PSU)	Phillips #1 and #2 Screwdrivers
SSD / M.2 NVMe SSD	Anti-Static Wrist Strap & Mat
Mid-Tower ATX Case	Tweezers and Cable Ties
CPU Cooler with Thermal Paste	Multimeter

Hardware Identification and Handling

Objective: Label parts, explain functions, use anti-static measures.

Background

The motherboard is the central hub connecting all PC components. It integrates the CPU socket, RAM slots, PCIe expansion slots, SATA ports, BIOS chip, and CMOS battery. The CPU executes instructions and is mounted in a socket (LGA for Intel, AM5 for AMD). RAM provides fast temporary storage for running programs. Storage devices (SATA SSD or NVMe M.2) hold data permanently. The PSU converts AC power to DC voltages for all components.

Anti-Static Safety Measures

- Wear an anti-static wrist strap connected to a grounded metal surface at all times.
- Work on an anti-static mat on a flat, non-carpeted surface.
- Handle all PCBs and cards on their edges only — never touch chips or pins.
- Store removed components in anti-static bags immediately.
- Touch the metal case of chassis before touching any component to discharge static.

Procedure

1. Put on an anti-static wrist strap and verify ground connection.
2. Arrange all components on the anti-static mat.

3. Identify each component and label the motherboard diagram below.
4. Describe the function of each component to the instructor.
5. Complete the identification record with make, model, and specifications.

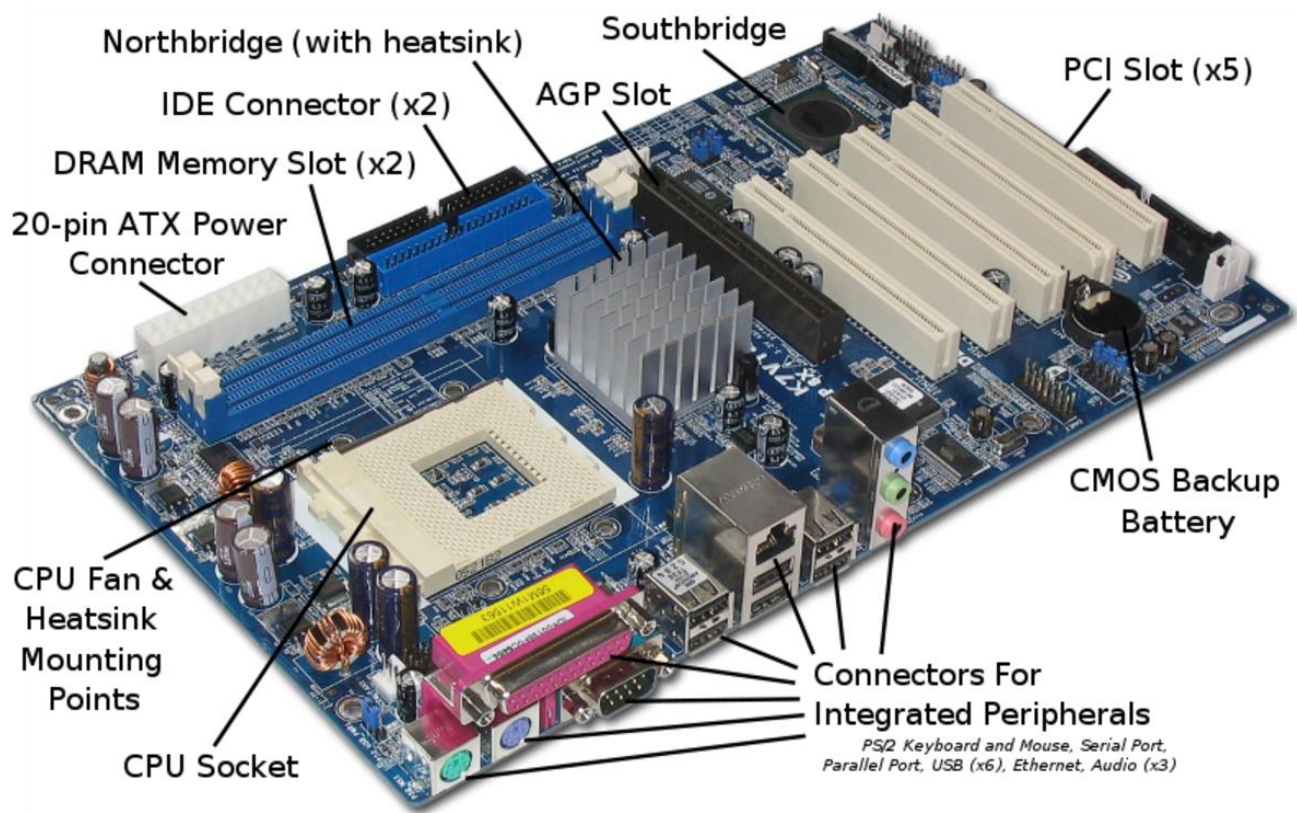


Figure 1: Motherboard Layout — Label the components on your worksheet

Background

PC assembly must follow a specific sequence to avoid blocking access to mounting points. Thermal paste is applied between the CPU and cooler to fill microscopic air gaps and improve heat transfer. A small pea-sized dot at the center of the CPU spreads to correct coverage when the cooler is pressed and secured. Improper assembly order or missing thermal paste leads to overheating or component damage.

Assembly Procedure

6. Install the I/O shield into the rear panel of the case.
7. Open CPU socket lever, align CPU (match the triangle marker), lower with zero force.
8. Apply a pea-sized amount of thermal paste to the center of the CPU.
9. Mount the CPU cooler, tighten screws in a diagonal pattern, and connect fan headers.
10. Insert RAM into dual channel slots (A2 and B2) until both clips click.
11. Install M.2 NVMe SSD and secure with the retention screw.
12. Mount motherboard on standoffs, align with I/O shield, secure with screws.
13. Connect 24-pin ATX power cable and 8-pin CPU EPS cable from PSU.
14. Connect SATA data and power cables to storage devices.
15. Connect front-panel headers (power button, reset, HDD LED, power LED).
16. Route and secure cables with zip ties.
17. Power on and verify POST: single short beep + BIOS display = success.

Disassembly Procedure

Perform a full shutdown, wait 30 seconds, then unplug the power cord before touching any component. Remove components in the reverse order of assembly. Place every removed component into a labeled anti-static bag immediately.

PC Assembly Procedure

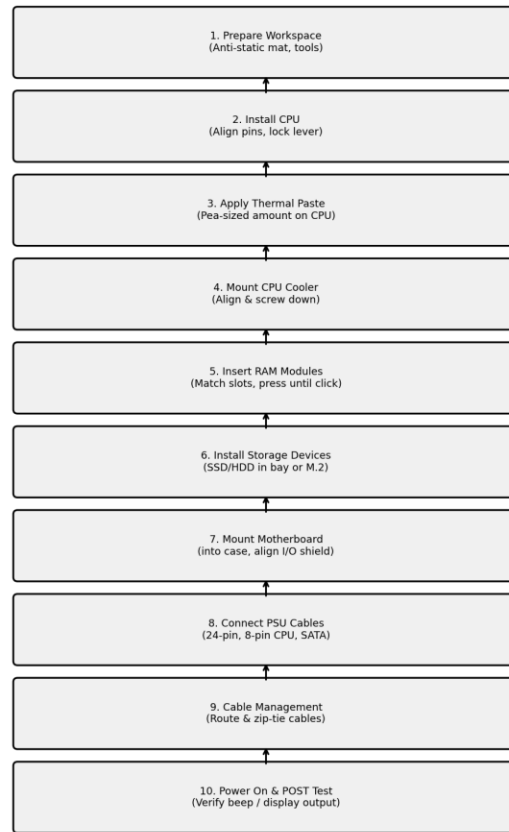


Figure 2: PC Assembly Step-by-Step Flowchart

Observations

Time to assemble (minutes):

POST result (beep / display output):

CPU temperature at idle (°C):

Errors or difficulties encountered:

Time to disassemble (minutes):

Section 9.3 — Network Setup (LAN Configuration)

Objective: Configure wired LAN, assign IPs, test with ping.

Background

A Local Area Network (LAN) connects devices within a limited area. This experiment uses a star topology where each PC connects to a central switch via a dedicated Ethernet cable. Star topology is the most common LAN design because a fault on one cable only affects that one device, leaving the rest of the network working.

IPv4 addresses are written in dotted decimal format (e.g., 192.168.1.1). The subnet mask (e.g., 255.255.255.0) defines which devices are on the same network. In this experiment, each PC is assigned a fixed (static) IP address manually, without a DHCP server.

Procedure

18. Connect each PC to the switch with a Cat5e cable. Confirm link LEDs are on.
19. Go to: Control Panel → Network and Sharing Center → Change Adapter Settings → Local Area Connection → Properties → IPv4 → Properties.
20. Select "Use the following IP address" and enter the assigned IP, subnet mask 255.255.255.0, and gateway 192.168.1.254.
21. Click OK to apply settings.
22. Run `ipconfig /all` in Command Prompt to verify the configuration.
23. Ping every other PC and record results in the table below.

Figure 3: Star-Topology LAN with IP Addressing Scheme

LAN IP Address Assignment Table

Device	IP Address	Subnet Mask	Default Gateway	Status
PC1	192.168.1.1	255.255.255.0	192.168.1.254	Active
PC2	192.168.1.2	255.255.255.0	192.168.1.254	Active
PC3	192.168.1.3	255.255.255.0	192.168.1.254	Active
PC4	192.168.1.4	255.255.255.0	192.168.1.254	Active
PC5	192.168.1.5	255.255.255.0	192.168.1.254	Active
PC6	192.168.1.6	255.255.255.0	192.168.1.254	Active
Router	192.168.1.254	255.255.255.0	—	Gateway

Figure 4: LAN IP Address Assignment Table

Connectivity Matrix — Record ping results (OK / FAIL)

From \ To	PC1	PC2	PC3	PC4	PC5	PC6
PC1	—					
PC2		—				
PC3			—			
PC4				—		
PC5					—	
PC6						—

Section 9.4 — Troubleshooting Simulations

Objective: Use POST card, interpret beep codes, fix connectivity issues.

Background

Systematic troubleshooting identifies and resolves faults step by step: (1) define the problem, (2) gather information, (3) identify possible causes, (4) test the most probable cause, (5) apply a fix, (6) verify resolution, (7) document the outcome. POST (Power-On Self-Test) runs automatically at startup and checks the CPU, RAM, video card, and storage. If a critical fault is found, POST signals it through beep codes.

POST Beep Code Reference

Beep Pattern	Indicated Fault	Corrective Action
1 Short	POST passed — normal boot	No action required
3 Long	RAM not detected	Reseat RAM; test one module at a time
1 Long, 2 Short	Video card fault	Reseat GPU; test with alternate card

Continuous long	RAM or PSU failure	Test PSU voltages; replace RAM
No beep, no display	CPU / PSU / mobo fault	Check PSU, CPU seating, all power cables

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Figure 5: POST and Troubleshooting Decision Flowchart

Troubleshooting Scenarios

For each scenario below, write what you observed, suspected cause, steps taken, and resolution.

Scenario A — System powers on, 3 long beeps, no display

Observation:

Suspected cause:

Steps taken:

Resolution:



Scenario B — PC boots but cannot ping any other device on the LAN

Observation:

Suspected cause:

Steps taken:

Resolution:

```
PC 1

Wireless LAN adapter Wi-Fi 2:

Connection-specific DNS Suffix . : 
IPv6 Address. . . . . : 2401:4900:1f2d:c54b:80b9:9a9e:b67d:b620
Temporary IPv6 Address. . . . . : 2401:4900:1f2d:c54b:394a:c5a6:21f0:4b27
Link-local IPv6 Address . . . . . : fe80::d5e0:d4d2:9e0b:a5a0%20
IPv4 Address. . . . . : 192.168.1.4
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::1%20
                          192.168.1.1

PS C:\Users\Admin\Desktop> ping 192.168.1.7

Pinging 192.168.1.7 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.7:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Scenario C — System on, no beeps, no display, fans at full speed

Observation:

Suspected cause:

Steps taken:

Resolution:



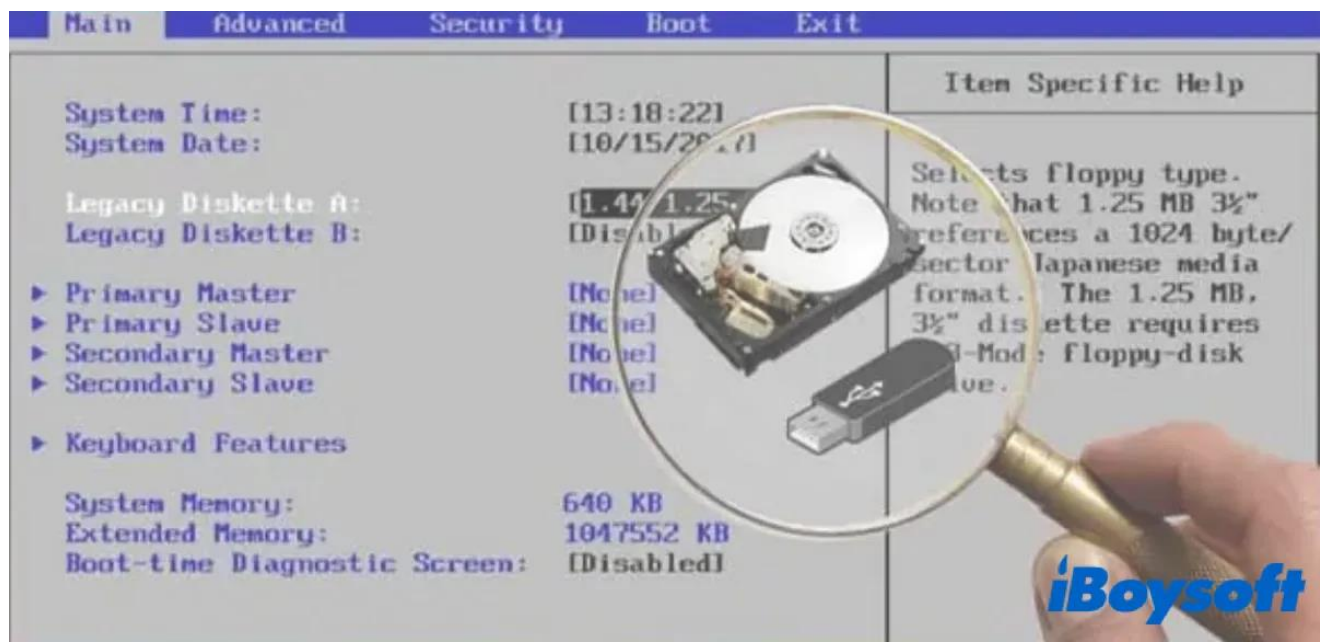
Scenario D — System boots normally but HDD not detected in BIOS

Observation:

Suspected cause:

Steps taken:

Resolution:



Discussion

The four experiments of Topic 9 show that hands-on practice is essential for building real competency in hardware and networking. Several important lessons came out of the work.

Key Findings

- ESD safety is critical. Modern components can be permanently destroyed by static discharge invisible to the eye. Wearing a wrist strap and using an anti-static mat must become habitual.
- Assembly order matters. Skipping steps, such as mounting the motherboard before attaching the CPU cooler, makes components inaccessible and forces partial disassembly to continue.
- Network faults can be silent. A wrong subnet mask caused complete connectivity failure with no error message. The PC appeared normal but could not reach any other host. Checking each network layer in order is the only reliable way to find this type of fault.
- Most hardware faults are simple. Loose cables, unseated RAM, and incorrect settings cause the majority of real-world PC failures. Always start with the simplest probable causes before moving to complex diagnostics.